Computer-aided Breast Cancer Diagnosis using Deep Convolutional Neural Networks

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***Abstract*- In the contemporary period, breast cancer has become one of the leading causes of cancer-related mortality globally. It is referred to as asymptomatic cancer because it has no symptoms at all when it is first diagnosed. As a result, radiology professionals periodically evaluate mammograms to check for unusual lesions and pinpoint the location, kind, and shape of any troublesome breast areas. For the segmentation and detection of breast cancers, image analysis techniques are vital because they can accurately depict key morphological features that are necessary for a conclusive diagnosis. This paper presents the image segmentation of a breast tumor using U-Net model. The proposed model attained a high accuracy of 94.29% on our training dataset. The complete empirical analysis along with the exhaustive literature review is presented in the paper.**

Keywords- Breast Cancer, U-Net, Semantic Image Segmentation

I. INTRODUCTION

Statistics show that the leading cause of death for women globally is breast cancer [1]. However, the majority of women are still unaware of this illness. They ignore or fail to recognize the symptoms. Though the mortality rate for men is significantly lower than that of women, it is exceedingly disturbing that both men and women have this malignancy. Fortunately, a lot of organizations are now stepping forward to inform people about the signs and remedies for this malignancy. Some organisations are even running campaigns and offering free check-ups to help detect the cancer sooner, which is highly appreciable.

We should also develop ways to identify this cancer's initial stages more precisely and its subsequent treatments which are more comfortable for the patients physically and mentally. Several people had conducted the study in this area and more people are coming forward to do the same.

II. LITERATURE SURVEY

A brand-new approach built on U-Net was put forth by Yu et al. [] to increase the accuracy of tumor segmentation in breast ultrasound pictures. They proposed to solve the vanishing gradient issue and introduced a new link, a thick block from the feature maps' input in the area that handles encoding and decoding, to lessen the feature information loss. Comparing the results of the suggested method's true-positive rate with those of the U-Net rate, false-positive rate, Jaccard similarity, Hausdorff distance indices, and Dice equations revealed a general improvement.

Convolutional neural networks (CNNs) were employed by Xu et al. [3] to segment three-dimensional (3D) breast ultrasound images into skin, mass, fibro glandular tissue, and fatty tissue. Compared to their earlier study utilizing the watershed technique, which had a 74.54 percent JSI, the JSI produces a result of 85.1 percent value.

Al-Masini et al. [4] proposed another deep learning model based on the You Only Look Once (YOLO) approach to identify and classify masses in digital mammograms. In order to eliminate artefacts from mammograms, the authors decide to use a multi-threshold peripheral equalization approach as a pre-processing step before the segmentation and classification procedures. The YOLO model is then given the pre-processed images to identify and categorize aberrant masses as benign or malignant.

Wang et al. described a technique for dividing breast tumour masses using a deep learning network [5]. They implemented an encoder and a decoder as part of a multi-level nested pyramid network. The dice coefficient was utilized by the authors to determine how well their suggested technique worked. However, extra metrics are required for the performance assessment of semantic segmentation to assess the effectiveness of the segmentation and classification phases.

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Ragab et al. [6] suggest a new computer-aided diagnosis (CAD) system based on the extraction of the features and classification utilising DL methods to help radiologists diagnose breast cancer lesions in mammography. When compared to other methods, the accuracy obtained utilizing deep features fusion for datasets was the greatest. On the other hand, the accuracy did not become better when the PCA was used on the feature fusion sets, but the computational price dropped as the execution time dropped.

A proposed architecture by Asma et al [7] uses additional modified skip connections to link two U-Nets. They also put the recommended design into practise on the Residual U-Net and the Attention U-Net. They show via their quantitative and qualitative analyses that the proposed design can result in a higher Dice score and better automated mass segmentation.

III. MATERIAL AND METHODS

A. Dataset Description

Breast ultrasound images taken among women between 25 and 75 are part of the baseline data. This data was gathered in 2018. There are 600 female patients in all. The collection comprises 780 images, each of which measures 500x500 pixels on average. The images are PNG files. This dataset has three sections: normal, benign, and malignant [8]. Fig. 1 shows a few illustrations of sample images.

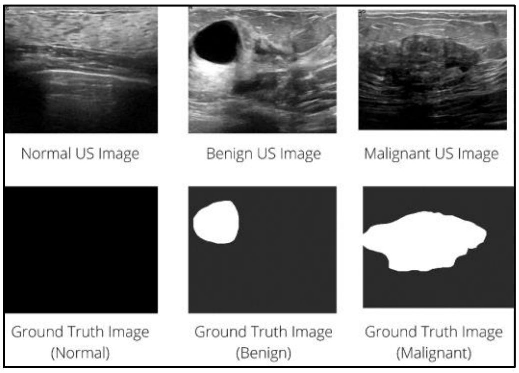


Fig. 1. Sample data images of Ultrasound Breast Images & Ground Truth Images

B. Proposed Framework

In this research, we presented semantic segmentation using the U-Net model. Fig. 2 depicts the flowchart of the process. The retrieved ultrasound images of breasts of our dataset were pre-processed into 256x256x3 pixels for the motive of training and testing. Over our dataset, we used Image Augmentation, which was then fed into a U-Net model for semantic image segmentation. The optimal weights based on reduced validation loss were achieved for the final modelling after this model had been trained for 50 epochs.

Breast ultrasound image data in the dimensions of 256x256x3 would be the input for this trained model. The U-Net model would then provide a segmented masked image.

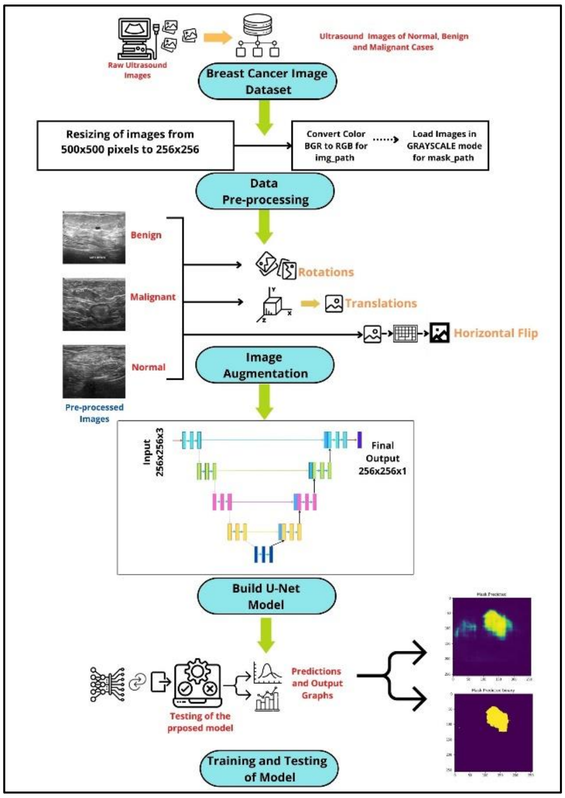


Fig. 2. Flow chart of the procedure of implementation

C. Image Augmentation

An efficient way to increase the amount of a dataset is to use the image augmentation technique. It is a method of altering original images by applying several transformations, producing numerous altered versions of the same image.

In order to prevent our algorithm from developing a fragile solution that will fail because it was only trained on a small image dataset, we augment our dataset so that every single new batch that is sent over to the algorithm sees a slight variation in the dataset every single time. Although the data will constantly change as a result, this will allow the algorithm to reach a more general solution and be able to generalize solutions and unseen data a little bit better.

For image augmentation, we have applied Keras Image Data Generator class. The input of the original data is obtained using the Keras Image Data Generator, which then transforms the input data randomly and outputs a result that solely contains the newly altered data. It does not increase the data. Using the Keras image data generator class, data augmentation is also done to broaden the generalisation of the model as a whole. With the use of an image data generator, data augmentation operations including rotations, translations, shear in, scale adjustments, and horizontal flips, are performed at random.

With the help of Image Data Generator, we created two distinct functions: one for normal images and another for masked images. An example outcome for each is shown in Fig. 3.

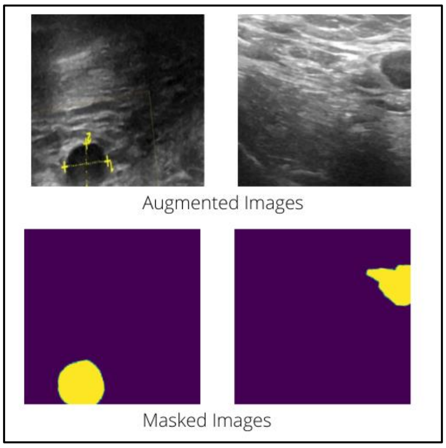


Fig. 3 Images Generated using ImageDataGenerator.

D. U-Net Semantic Image Segmentation

We have used U-Net architecture in our research to semantically segment breast ultrasound images. Each pixel in an image is given a class that specifies what it is meant to represent when semantic image segmentation is used. This process is frequently referred to as dense prediction since we make predictions for every pixel in the image.

U-Net model is proposed for Biomedical Image Segmentation [9]. This model performs semantic segmentation. The first path is the Contraction path (Encoder), which uses stacked convolutional layers and maxpooling layers to understand the context of the medical image. With the help of transposed convolutions, the second path, known as the symmetric Expanding path (Decoder), enables exact localization.

Fig. 4 shows a block schematic of the U-Net model. This model has 3x3 convolution layers and 2x2 maxpooling layers in the blocks of the contraction(encoder) phase. While the depth gradually increases from 256x256x3 to 16x16x256, the size of the images gradually decreases in the contraction path. After the encoder layers, there are 3x3 convolution layers and 2x2 transposed convolution levels, which results in expanding layers. Transposed convolutions are used in the expansion(decoder) path in addition to regular convolutions. From 16x16x256 to 256x256x1, the image size here gradually grows as the depth gradually lowers.

Our U-Net model's hyperparameters include a 3x3 kernel size, the activation function ReLU (Rectified Linear Unit), the kernel initializer "he_normal," and the padding "same." After each epoch, the model performance was improved using the Adam optimizer function with a learning rate of 0.00001.

We employed the BCE, Dice coefficient, and Dice loss functions. The predicted segmentation's pixel-by-pixel agreement with the associated ground truth can be compared using the Dice coefficient.

The Dice Coefficient is a well-liked evaluation statistic for tasks based on image segmentation that assesses sample overlap. It is equal to two times the overlapping area divided by all of the pixels in the images. The formula is given by:

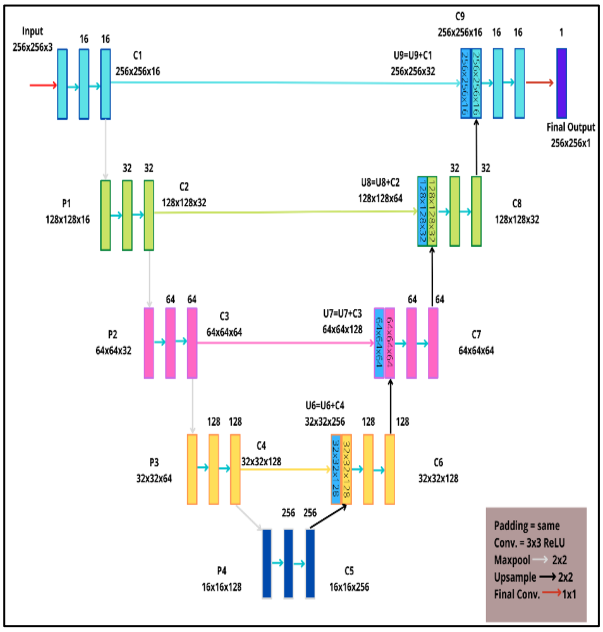


Fig. 4. Schematic Diagram of U-Net Architecture

$$\frac{2|A\cap B|}{|A|+|B|} \quad (1)$$

$|A\cap B|$ is ‘A’ intersection ‘B’ which shows the shared elements between sets A and B. The loss for our U-Net model at each epoch is computed by considering Eq. 1.

In the field of computer vision, the Dice coefficient is a popular statistic for determining how similar two images are. It has been modified into a loss function called Dice Loss. Eq. (2) states the formula of dice loss [10].

$$DL(y,\hat{p}) = 1 - \frac{2y + \hat{p} + 1}{y + \hat{p} + 1} \quad (2)$$

In order to lower the mean probability error between the target and predicted label for each pixel, BCE (Binary Cross Entropy) is used as the training method [11].

$$Logloss = -\frac{1}{N}\sum_{i=1}^{N} y_i \log(p(y_i)) + (1 - y_i)\log$$
(3)

In our U-Net model, Eq. 3 is applied to calculate loss at each epoch.

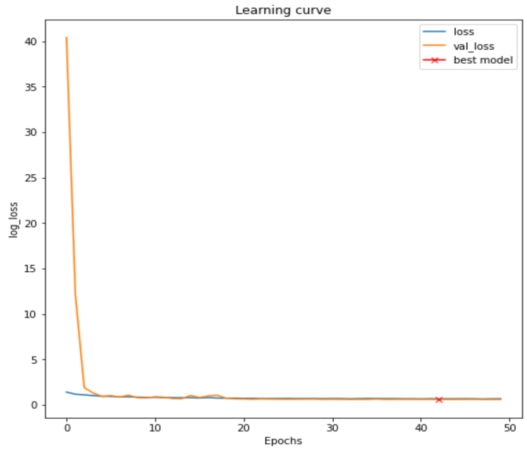
IV. RESULT AND DISCUSSION

Below table 1 displays the results of U-Net segmentation on the testing dataset in regards to accuracy, Dice Coefficient Score, Dice Loss, and Binary Cross Entropy Loss.

TABLE 1: PERFORMANCE EVALUATION SCORE OF IMAGE SEGMENTATION

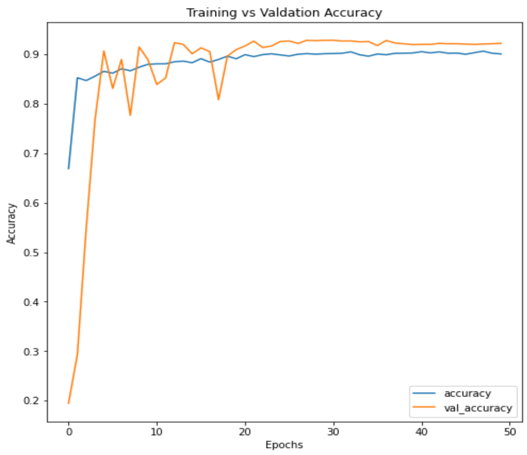
BCE loss/log loss	0.5998
Dice Loss	0.4215
Dice Coefficient	0.5785
Accuracy	0.9429

Fig. 5 depicts the U-NET model's learning curves. It illustrates how model performance improves as the number of epochs increases. For training and validation images, efficiency is improving concerning loss, accuracy, and dice coefficient score. The proposed U-Net model's loss is shown in Fig.5(a), which is less than one after 50 epochs.

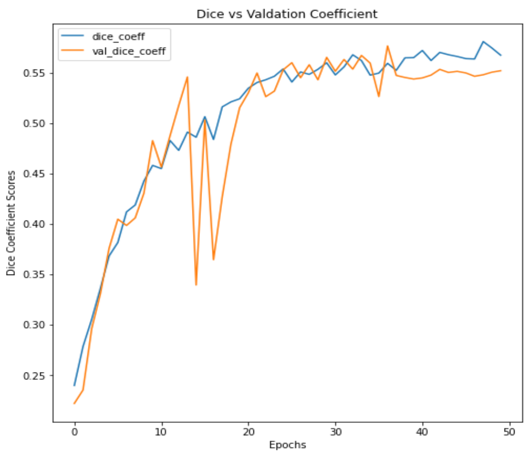


5(a) Learning curve for U-Net on Binary Cross Entropy Loss

Fig. 5(b) provides an illustration of the proposed method's accuracy. This variable explains how well the model performs across all classes in general. According to the results, high accuracy (0.9429) was attained after 43 epochs.



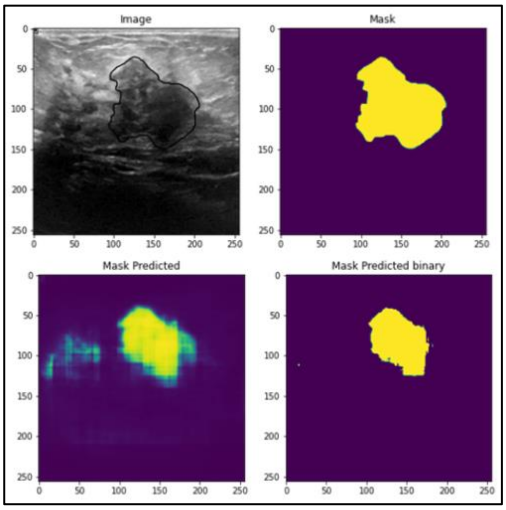
5(b) Learning curve for U-Net on Accuracy



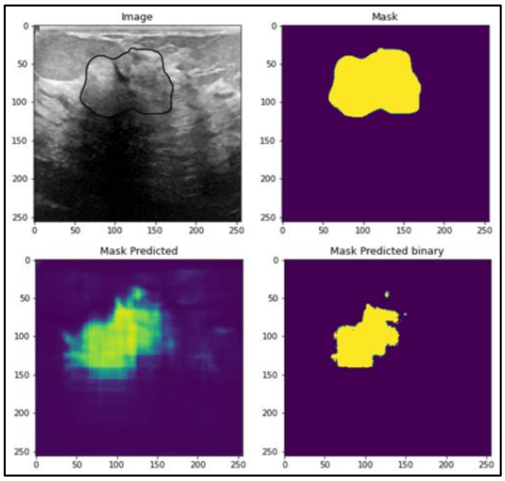
5(c) Learning curve for U-Net on Dice Coefficient

Fig. 5. Semantic Image Segmentation with U-NET model

In Fig.5(c), the dice coefficient is depicted. The dice coefficient is employed and derived as in Eq. (2) for model evaluation on dice loss, where A is the anticipated collection of pixels and B is the ground truth.



(a)



(b)

Fig. 6. Input Image, Masked Image, Mask Predicted, and Mask Predicted Binary of the proposed U-NET model

Fig. 6, which illustrates the ultrasound image dataset's input image and predicted images, provides an illustration of the experimental findings. The segmentation of a breast tumour utilizing the U-net model architecture is shown in this study. This architecture attains adequate performance in the segmentation of breast ultrasound images.

Table 2 shows the comparative analysis if evaluation metrices performed by the U-Net model with our proposed U-Net model. We can see we have achieved greater accuracy compared to the other two models.

TABLE II COMPARATIVE ANALYSIS OF EVALUATION METRICES

Paper Title	**Dataset**	**Metrices**	**Metrices Values**

Breast cancer mitotic cell detection using cascade convolutional neural network with U-Net [12]	ICPR2014 Dataset	Mean Dice Coefficient Accuracy Recall F-Score Precision	0.715 0.726 0.694 0.571 0.485
Breast Tumour Segmentation Using Deep Learning by U-Net Network [13]	Marginalised Dataset from TCGA	Accuracy Precision Recall	92.50 92.00 90.00
Computer-aided Breast Cancer Diagnosis using Deep Convolutional Neural Networks [proposed]	Breast Ultrasound Images	BCE Loss Dice Loss Dice Coefficient Accuracy	0.5998 0.4215 0.5785 0.9429

V. CONCLUSION

Breast cancer is a complex disease suffered by many women in the world which not only harms their physical health but also takes a toll on their mental health. Not only women but also men can suffer from this disease. When a woman develops this cancer, not only does she suffer but her entire family does as well. One of a woman's most vital body parts is her breast, which she may be recommended to remove during therapy, leaving them broken within. Therefore, a timely diagnosis of this illness is essential for subsequent therapy.

Since many individuals are unaware of this, they are unable to recognize the symptoms of the disease, but many organizations are taking the initiative to raise awareness of breast cancer and conducting free check-ups, which is already a good step in this direction. Also, more people are coming forward to research the methods to detect and treat at preliminary stage, and we also tried to do the same.

In this study, we looked at how computer vision and deep learning methods may fully automate the early diagnosis of breast cancer using ultrasound images. For the same, we used the U-Net model, which classifies the infected area in the images through image segmentation. The data set we used contains 437 images of benign tumours, 210 images of malignant tumours, and a total of 647 images of 500x500 pixels. In this research, we have used Semantic Image Segmentation using the U-Net model. We passed our dataset in our proposed model and attained a high accuracy of 94.29% in our training dataset, along with less loss in the outcome.

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